House price predictor model comparison:

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Methodology:

This project aims to predict house prices using the California Housing dataset provided by Scikit-Learn. The dataset contains 20,640 records with 8 numerical features such as median, income, house age, average rooms, and geographic location.

First the dataset was loaded using the fetch_california_housing() function and converted into a pandas DataFrame. The target variable was MedHouseVal, representing the median house value, while the remaining variables were used as input features.

Next, feature scaling was applied using StandardScaler to normalize the data and improve model performance. The dataset was then split into training and testing sets using an 80:20 ratio.

Three regression models were trained:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Results

The performance of the models was evaluated using RMSE (Root Mean Squared Error) and R^2 Score.

Model	RMSE	R^2 Score
Linear Regression	0.745	0.575
Ridge Regression	0.745	0.575
Decision Tree	0.724	0.599

Conclusion:

This project demonstrated how machine learning models can be used to predict house prices based on multiple features. Feature scaling improved the performance and stability of the models.

Among the models tested, the Decision Tree Regressor provided better performance compared to Linear and Ridge Regression.